

Econometrics

Problem Set 4 (due 12/10 at 11:59 p.m.)

1. This question discusses RDD with a non-binary treatment and is based on Caetano, Caetano and Escanciano (2023), which can be found on eLC. We will examine the differences between the standard RDD, which assumes a binary treatment, and multi-valued treatment, which is very common in applications.

The dataset MedicarePS is a reduced version of the original dataset with only a small subset of variables. Because of this, you should not expect to find the exact same results as in the paper. However, reading over the paper may be helpful.

Use the command “describe” to understand the data. The outcome is DelayCare, which is an indicator for whether the person delayed care in the previous year due to cost reasons. The treatment in which we are interested is insurance. The treatment variable Ninsurance is not binary, since it can assume the values 0, 1 and 2 depending on how many policies the person has. The other variables should be self-explanatory.

- (a) At age 65 people become eligible to Medicare. Explain why this may be a good setting for the RDD with Age as the running variable and 65 as the threshold. Discuss the plausibility of the validity, relevance and monotonicity conditions.
- (b) Do 3 plots showing the averages per quarter of age: outcome, first stage and validity.
 - i. Does it seem like relevance holds?
 - ii. Does the validity plot support the validity assumption?

- (c) Since the treatment is not binary, we will need to make assumptions in order to estimate treatment effects. Specifically, we assume that the model is

$$DelayCare_i = \beta_0 + \beta_1 Ninsurance + U_i.$$

This is what is normally done in the RDD with a regular (non-binary) variable, though most of the time the model above is assumed implicitly. The model is then estimated identically to the standard RDD, using the Wald estimator $\hat{\theta}_{FRD}$ in the notes.

We will estimate this model using the 2SLS approach.

- i. Create a variable Z which is an indicator for whether the person is at least 65 years old, and a variable $AgeZ$ which interacts Age with Z .
 - ii. Program a triangular kernel function, which we will call k .
 - iii. Drop the observations that are younger than 60 and older than 70 (this is equivalent to using a bandwidth $h = 5$).
 - iv. Run a 2SLS regression of $DelayCare$ on $Ninsurance$, using Z as an IV and Age and $AgeZ$ as exogenous controls, and k as weights. Use the “robust” option inside the “ivregress” (this is a Stata command, do the analogous for R). What is the effect of the number of insurances on the likelihood that one delays care because of cost reasons? Is this result significant at standard levels?
- (d) Repeat the procedure in the previous question for $h = 3$ and $h = 7$. Do the results change? Looking at the outcome plots, is this coherent with what you would expect given the curvatures of these functions?
- (e) If the standard assumptions of the RDD hold, we don’t need to use covariates. However, Calonico, Cattaneo, Farrell, and Titiunik (2019) defend the use of covariates to improve the efficiency of the estimation. Run the RDD adding the

covariates Minority and Education as exogenous controls. Does this generate a major effect in the main estimate? Does the variance decrease?

2. Suppose you don't want to assume the linear model of the previous question. What if you want to estimate separately the effects of going from zero insurances to one insurance, and from one insurance to two insurances?

- (a) Is the linear model of the previous question reasonable? Why may someone want to allow for heterogeneous marginal effects instead?
- (b) Create two dummy variables, *Oneins* and *Twains*. The first is equal to one if the person has at least one insurance, and the second is equal to one if the person has at least two insurances.
- (c) The model we want to estimate is

$$DelayCare_i = \beta_0 + \beta_{1i}Oneins + \beta_{2i}Twains + U_i. \quad (1)$$

Why can't we simply run the regression in item c(iv) in the previous question using *Oneins* and *Twains* instead of *Ninsurance*?

- (d) Reproduce Figures 2 (a) and (b) in Caetano, Caetano and Escanciano (2023). The plot on the left is done using only whites (Minority=0). The plot in the right is done using only minorities (Minority=1).
- (e) In the plots of the previous item, note how the first-stages of *Oneins* and *Twains* are completely different for different races. Let δ_{1Wh} be the discontinuity at 65 in the probability of being insured at all among whites, and δ_{1Min} the analogous for the minorities. Let δ_{2Wh} be the discontinuity at 65 in the probability of having 2 insurances among whites, and δ_{2Min} the analogous for the minorities. Eyeball those values and show that $(\delta_{1Wh}, \delta_{2Wh})'$ and $(\delta_{1Min}, \delta_{2Min})'$ are linearly independent.

(f) Do one Intention to Treat (i.e., reduced-form) figure showing the average outcome for whites for different values of the running variable (age), and another analogous figure for minorities. The discontinuity in the outcome for whites reflects the effect of the discontinuity in both variables Oneins and Twins at the same time. The same is true for minorities. In principle the effects of these two discontinuities cannot be disentangled.

(g) Using model (1), write the equation for the discontinuity in the outcome among whites

$$\lim_{a \downarrow 65} E[\text{DelayCare} | \text{Age} = a, \text{Minority} = 0] - \lim_{a \uparrow 65} E[\text{DelayCare} | \text{Age} = a, \text{Minority} = 0] = ?$$

as a function of the δ 's from item (e). Now do the same for minorities. This gives us a system. In order to transform this into a solvable system we need three assumptions. What are they? (Hint: one assumption is about the unobservables, another is about the treatment effects, and another is about being able to invert the system to find a solution).

(h) Eyeball the discontinuities in the outcomes in the plots of item (f). Substitute these and the eyeballed δ 's into the system and solve it. What are the treatment effects?

(i) Now let's do this properly.

- i. Create some interaction variables: AgeMin (Age and minority interacted), AgeZ, MinZ and AgeMinZ ("xxxZ" means that we interact the variable named "xxx" with Z).
- ii. Program k and restrict the observations within bandwidth distance as in the previous question.
- iii. Run a 2SLS regression of DelayCare on Oneins and Twins, using Z and MinZ as IVs and Age, AgeZ, Minority, AgeMin and AgeMinZ as exogenous

controls, and k as weights. Use the “robust” option. What are the treatment effects? Are they similar to the results you eyeballed? Are they significant? What is the main driver of the decision of delaying care due to cost reasons: having insurance at all, or being better covered by two policies?

- (j) We generated a 2×2 system using *Minority*. We also have the variable *Education*, which takes 3 values. You could have created a 3×2 system, looking at high school dropouts, those with (exactly) high school completed and those with at least some college separately. For example:

$$\lim_{a \downarrow 65} E[\text{DelayCare} | \text{Age} = a, \text{Education} = 1] - \lim_{a \uparrow 65} E[\text{DelayCare} | \text{Age} = a, \text{Education} = 1] = ?$$

and the same for *Education*=2 and *Education*=3.

You could also create a 6×2 system by looking at each combination of race and education. For example:

$$\begin{aligned} \lim_{a \downarrow 65} E[\text{DelayCare} | \text{Age} = a, \text{Minority} = 0, \text{Education} = 1] \\ - \lim_{a \uparrow 65} E[\text{DelayCare} | \text{Age} = a, \text{Minority} = 0, \text{Education} = 1] = ? \end{aligned}$$

- i. Suppose that there are homogeneous treatment effects. In order to identify the treatment effects, what other assumptions are necessary? (Hint: same kind of reasoning as in item (g).) Are the assumptions different from item (g)? Weaker or stronger?
 - ii. What else can we identify using these 6 equations? Can we relax the assumption of homogeneous treatment effects?
3. Suppose that we just want to identify the effect of *Oneins* and *Twins* assuming homogeneous treatment effects. Although we showed in item (g) and even did it in item (i) using only *Minority*, we can also do it with the *Minority* $\times$ *Education* combinations.
- (a) Create 5 dummies, one for each combination of race and education. Omit the

minority high school dropout combination. Call this vector Q .

- (b) Create some interaction variables: AgeQ (Age interacted with each element of Q), AgeZ, QZ and AgeQZ.
- (c) Program k and restrict the observations within bandwidth distance as in the previous questions.
- (d) Run a 2SLS regression of DelayCare on Oneins and Twins, using Z and QZ as IVs and Age, AgeZ, Q, AgeQ and AgeQZ as exogenous controls, and k as weights. Use the “robust” command. What are the treatment effects? Are they similar to the effects in (i)? If they are different, and in light of the different assumptions of both methods as discussed in item (j)i, does the result make sense? What about the standard errors? (Note that using more variables tends to substantially decrease the standard errors, since we are using more instruments and are thus over-identified. In this particular example we have 4 over-identifying restrictions.)